

Problem 1

As usual, assume that $\delta(t)$ and $u(t)$ denote the Dirac function and the Heaviside function, respectively.

Use the Laplace transform to solve the initial value problem

$$y''(t) + 9y(t) = \delta(t-1) + u(t-1), \quad t > 0$$

with initial conditions $y(0) = y'(0) = 0$.

$$y''(t) + 9y(t) = \delta(t-1) + u(t-1), \quad y(0) = y'(0) = 0$$

$$s^2 F(s) + s F(0) + f'(0) + 9 F(s) = e^{-s} + \frac{e^{-s}}{s}$$

$$F(s) (s^2 + 9) = e^{-s} (1 + \frac{1}{s})$$

$$F(s) = e^{-s} (\frac{1}{s^2 + 9} + \frac{1}{s(s^2 + 9)})$$

Delbrotk.:

$$\frac{1}{s(s^2 + 9)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 9} \Rightarrow 1 = A(s^2 + 9) + s(Bs + C) = As^2 + 9A + Bs^2 + Cs \Rightarrow \begin{matrix} A+B=0 \\ C=0 \\ 9A=1 \end{matrix} \Rightarrow A = -B = 1/9, C=0$$

$$\mathcal{L}^{-1}(F(s)) = \frac{e^{-s}}{9} (\frac{1}{s} - \frac{s}{s^2 + 9}) = \frac{1}{9} u(t-1) (1 - \cos(3(t-1)))$$

Problem 2

Again, let u be the Heaviside function.

a) Sketch the function $-u(x + \pi) + 2u(x)$ over the interval $[-\pi, \pi]$.

b) Compute the Fourier series $f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$ for

$$\text{the function } f(x) = \cos 3x - \frac{1}{2} \sin 2x - u(x + \pi) + 2u(x).$$

$$a) \quad x = -\pi : -u(0) + 2u(-\pi) = -1 + 2 \cdot 0 = -1$$

$$x = 0 : -u(0) + 2u(0) = -1 + 2 \cdot 1 = 1$$

$$x = \pi : -u(2\pi) + 2u(\pi) = -1 + 2 \cdot 1 = 1$$



$$b) \quad f(x) = \cos(3x) - \frac{1}{2} \sin(2x) + \underbrace{(-u(x + \pi) + 2u(x))}_{f_2}$$

$$f_2 : 0000 \Rightarrow a_0 = a_1 = 0$$

$$\cos(n\pi) = (-1)^n$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) dx = \frac{2}{\pi} \int_0^{\pi} \sin(nx) dx = -\frac{2}{\pi n} [\cos(nx)]_0^{\pi} = -\frac{2}{\pi n} (\cos(n\pi) - 1) = \frac{2 - (-1)^n}{\pi n}$$

$$f(x) = \cos(3x) - \frac{1}{2} \sin(2x) + \sum_{n=1}^{\infty} \frac{2 - (-1)^n}{\pi n}$$

Problem 3

Given is the heat equation

$$\partial_t u(x, t) = \partial_{xx} u(x, t), \quad 0 \leq x \leq 4,$$

together with initial condition

$$u(x, 0) = 2 \sin(3\pi x)$$

and boundary conditions

$$u(0, t) = u(4, t) = 0, \quad t > 0. \Rightarrow \text{Dirichlet}$$

a) Find all solutions of the heat equation which are of the form

$$u(x, t) = F(x)G(t) \text{ and also satisfy the boundary conditions.}$$

b) Find the solution u to the heat equation which satisfies both the boundary and the initial conditions.

$$a) \quad u_t(x, t) = u_{xx}(x, t), \quad 0 \leq x \leq 4$$

Suaret er på formen:

$$u(x, t) = F(x)G(t)$$

setter det inn i varmeledning:

$$F(x)G'(t) = F''(x)G(t)$$

Flytter og bytter:

$$\frac{F''(x)}{F(x)} = \frac{G'(t)}{G(t)} = k \quad \text{Må være lik en konstant } k.$$

ser at vi har to separable differensialligninger

$$\frac{F''(x)}{F(x)} = k \quad \wedge \quad \frac{G'(t)}{G(t)} = k$$

$$K = -p^2$$

$$F''(x) + p^2 F(x) = 0 \Rightarrow F(x) = A \cos(px) + B \sin(px), \quad p = \frac{\pi}{L}$$

sjekker randbet.

$$F(0) = A \cos(0) = 0 \Rightarrow A = 0$$

$$F(4) = B \sin(4p) = 0 \Leftrightarrow p = \frac{4\pi}{a}, \quad B \neq 0$$

$$F_n(x) = B \sin\left(\frac{n\pi x}{4}\right)$$

$$G'(t) + p^2 G(t) = 0 \Rightarrow G(t) = G_0 e^{-\left(\frac{c\pi n}{4}\right)^2 t}$$

$$u_n(x, t) = B_n \sin\left(\frac{n\pi x}{4}\right) e^{-\left(\frac{c\pi n}{4}\right)^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{4}\right) e^{-\left(\frac{c\pi n}{4}\right)^2 t}$$

Bestemmer B_n uha initialbet.

$$F(x) = u(x, 0) = \sum_{n=1}^{\infty} G_0 \sin\left(\frac{n\pi x}{4}\right)$$

Fourierkoeff. til $F(x)$ gir

$$B_n = \frac{2}{L} \int_0^L F(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$u(x, 0) = 2 \sin(3\pi x)$$

$$B_n = 2, \quad n = 3\pi$$

$$u(x, t) =$$

Problem 4

Given is the wave equation on the real line

$$\partial_{tt} u(x, t) = 16 \partial_{xx} u(x, t), \quad x \in \mathbb{R}, t > 0, \Rightarrow c = 4$$

with initial conditions

$$u(x, 0) = x, \quad \partial_t u(x, 0) = \cos(x). \Rightarrow \text{d'Alemberts}$$

Find the solution to this problem, and simplify the resulting expression as much as possible.

$$\begin{aligned} u(x, t) &= \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x} \\ &= \frac{1}{2} (x+4t) + (x-4t) + \frac{1}{2 \cdot 4} \int_{x-4t}^{x+4t} \cos(\tilde{x}) d\tilde{x} \\ &= x + \frac{1}{8} [\sin(\tilde{x})]_{x-4t}^{x+4t} \\ &= x + \frac{1}{8} (\sin(x+4t) - \sin(x-4t)) \\ &= \underline{x + \frac{1}{4} \cos(x) \sin(4t)} \end{aligned}$$

2 Fixed-point iteration

Given the function $g(x) = \frac{e^x}{10}$. Does this function have a fixed-point in the interval $[-1, 1]$? And if so, is this fixed-point unique? Explain your answers.

Fill in your answer here

The function has a fixed-point in the interval $[-1, 1]$ if $g(x)$ is continuous on the interval $[a, b]$, and $g(x) \in [a, b] \forall x \in [a, b]$

Since $-1 < \frac{e^{-1}}{10} < \frac{e^1}{10} < 1$ and e^x is monotone, so $-1 < \frac{e^x}{10} < 1$ for $x \in [-1, 1]$ there exists a f-p.

The fixed-point is unique if $g'(x)$ exists $\forall x \in [-1, 1]$, and there is a pos. constant L s.t. $g'(x) \leq L < 1 \forall x \in [a, b]$

The f - p is unique, since $|g'(x)| = \left| \frac{e^x}{10} \right| < e^1 < 1$ on $[-1, 1]$

since $g'(x) = g(x) > 0$ and $g'(x)$ is monotone.

3 Compute interpolation polynomial.

Given the points $x_0 = 0.0, x_1 = 1.0, x_2 = 2.0, x_3 = 4.0$ together with 4 data points

$f_0 = 1, f_1 = 5, f_2 = 10, f_3 = 12$. Find the polynomial of minimal degree interpolating through these points. Express it in the form

$$p_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

Only state the final solution, do not give a detailed derivation.

x_i	0	1	2	4
y_i	1	5	10	12

$$[y_0] = 1 \quad [y_1] = 5 \quad [y_2] = 10 \quad [y_3] = 12$$

$$[y_0, y_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{5-1}{1-0} = 4 \quad [y_1, y_2] = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10-5}{2-1} = 5 \quad [y_2, y_3] = \frac{y_3 - y_2}{x_3 - x_2} = \frac{12-10}{4-2} = 1$$

$$[y_0, y_1, y_2] = \frac{5-4}{2-0} = \frac{1}{2} \quad [y_1, y_2, y_3] = \frac{1-5}{4-1} = -\frac{4}{3}$$

$$[y_0, y_1, y_2, y_3] = \frac{\frac{-\frac{4}{3} - \frac{1}{2}}{4-0}}{\frac{\frac{1}{2}(x-1)(x-0)}{x^3 - 3x^2 + 2x}} = -\frac{44}{6} = -\frac{22}{3}$$

$$N(x) = 1 + 4x + \frac{x^2}{2} - \frac{x}{2} - \frac{22x^3 + 22x^2 - 56x}{x^3 - 3x^2 + 2x}$$